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SUMMARY

Software engineer with 3+ years of industry experience and an MS in Software Engineering (ASU, May 2026). Currently building AI-native, RAG-powered compliance automation at Pretorin; previously shipped data/ETL infrastructure at Amazon Web Services and founded the full technical stack of a consumer fintech app. Proficient in Python, TypeScript, and SQL, with deep experience in LLM/RAG systems (pgvector, OpenAI SDK), cloud platforms (AWS, GCP), PostgreSQL, and production-grade full-stack development. AWS Certified Solutions Architect – Associate.

WORK EXPERIENCE

Pretorin

May 2026 – Present

Product Engineer Intern

Tempe, AZ

- Contributing to an AI-native GRC platform that uses RAG workflows to automate evidence retrieval, compliance documentation, and control-readiness tasks for frameworks like FedRAMP, NIST 800-53, SOC 2, CMMC Level 2, and ATO.
- Supporting AI pipelines where uploaded evidence (policies, logs, screenshots, architecture docs) is extracted, chunked, embedded, and semantically retrieved via PostgreSQL + pgvector, then passed to LLMs to generate grounded compliance narratives with human review and audit traceability.
- Working across a stack of React, FastAPI, PostgreSQL, pgvector, Docker, and the OpenAI SDK to ship compliance automation features.
- Helping improve control-to-evidence mapping, AI-generated documentation, human review flows, and audit traceability; raising product quality through bug fixes, testing, and validation of AI-assisted workflows.

TappedIn

Nov 2025 – May 2026

Technical Co-founder

Remote, US

- Built the full technical stack (zero to TestFlight in ~5 weeks) for a fantasy market where users invest virtual currency in music artists: React Native + Expo frontend, Supabase backend with 28 migrations and 23 RLS policies.
- Architected an atomic trading engine with FOR UPDATE locks and idempotency keys; single PostgreSQL transactions prevent overdraft races and maintain an append-only audit trail.
- Designed queue-based batch pricing (pgmq) grouping 50 artists per Spotify API call with failure isolation and exponential backoff — a 50x reduction in external API calls.
- Shipped 11 screens, 22 components, 7 Supabase Edge Functions, and 12 RPCs; wrote ~950 test cases across Jest, Vitest, and pgTAP (full RLS validation); validated demand with 44 beta testers.

Handshake AI Solutions, LLC

Sep 2025 – Oct 2025

AI Trainer – Computer Science Expert

Remote, US

- Developed and evaluated 50+ domain-specific prompts to assess LLM performance across specialized CS subfields, covering edge-case scenarios and nuanced domain knowledge.
- Reviewed 100+ LLM outputs for accuracy, clarity, and depth; structured feedback improved internal accuracy metrics by 12% and reduced factual errors by ~20% in test runs.
- Designed new evaluation categories through independent literature research, expanding test coverage by 30% and accelerating the model fine-tuning feedback cycle.

Amazon Web Services (AWS)

May 2025 – Aug 2025

Software Development Engineer Intern

Seattle, WA

- Developed an AWS Glue ETL pipeline in Scala & TypeScript to replace a 3+ hr manual KDS migration SOP, transforming JSON/DynamoDB data into partitioned Parquet for Athena queries; cut prep time by ~95%.
- Benchmarked Glue vs. Step Functions for ETL and Redshift vs. Athena+S3 for storage on 500 GB test data; authored a cost/performance design doc adopted by a 4+ engineer team.
- Authored optimized SQL over partitioned Parquet outputs, enabling real-time migration assessments across multi-region datasets (1+ TB) and ad-hoc analytics.
- Productionized PySpark prototypes into Scala, right-sized compute, added a custom bookmarking solution, and fixed crawler race conditions; reduced runtime by 50%.
- Provisioned infrastructure via CDK (TypeScript) and CloudFormation (S3, Glue, Athena, IAM); built QuickSight dashboards and 10+ CloudWatch alarms monitoring 100+ active streams, resolving 95% of incidents within SLA.

o9 Solutions, Inc.

Jul 2022 – Jul 2024

Software Development Engineer

Bengaluru, India

- Delivered 10+ reusable modules in a modular tenant architecture using .NET (C#), MSSQL, and JavaScript/jQuery, improving scalability and configuration management across projects.
- Created a Kendo UI conflict-resolution grid that reduced merge time and boosted workflow efficiency for 20+ client-facing teams by 40%.
- Boosted test coverage by 30% with new .NET and JavaScript unit tests; reinforced Jasmine/MSTest suites with mocked payloads and regression cases.
- Resolved 250+ bugs across C# and JavaScript; refactored 60% of the test codebase, cutting duplication by 10K+ LOC and improving long-term maintainability.

OpenNets

May 2021 – May 2022

Software Developer

Bengaluru, India

- Architected a network topology simulator using Mininet, Node.js, and MongoDB supporting 10+ distinct topologies; automated checks cut configuration-related bugs by over 50%.
- Built an AngularJS configuration UI to visualize and manage topologies, cutting setup time per topology by ~40%.
- Enabled immediate on-call notifications via operational dashboards and alerting; resolved 95% of incidents within SLA.

EDUCATION

Arizona State University, Tempe, AZ

Aug 2024 – May 2026

Master's in Software Engineering (Computer Science) | CGPA: 3.67/4

Coursework: Statistical Machine Learning, Deep Learning, Advanced DSA, Software Design/Requirements/Testing

National Institute of Technology Karnataka, Surathkal, India

Jul 2018 – May 2022

Bachelor of Technology in Electronics and Communications Engineering

Coursework: Crypto and Blockchain Technologies, Image and Video Processing, Speech and Audio Processing

TECHNICAL SKILLS

Languages: Python, SQL, TypeScript, JavaScript, Scala, C#, C++, HTML, CSS

AI / ML: RAG pipelines, pgvector, OpenAI SDK, LLM evaluation, TensorFlow, PyTorch, scikit-learn, Pandas, NumPy, OpenCV, NLTK

Frameworks: React, React Native (Expo), Next.js, FastAPI, Node.js, Flask, .NET, Angular

Cloud / Databases: AWS (Glue, Athena, S3, CDK, SageMaker), GCP Cloud Run, Azure, PostgreSQL, Supabase, MongoDB, MSSQL, Oracle, Qdrant

Tools: Docker, Git, CI/CD, JIRA, Jest, Vitest, pgTAP, Jasmine/MSTest

Certifications: AWS Certified Solutions Architect – Associate

PROJECTS

FluidFill — AI Document Automation

Nov 2025

Python, FastAPI, Next.js, Postgres, Gemini API

- Built an AI tool turning static .docx legal templates into interactive forms: an Express gateway streams uploads to Postgres while fanning out parallel FastAPI jobs (placeholder parsing, HTML preview); Gemini returns a typed schema with a deterministic fallback so the flow never stalls; reconstructs pixel-faithful, styled DOCX on download.

Realtime Study Rooms

Oct 2025

React, Node.js, WebSockets, WebRTC, TypeScript

- Designed a Node.js signaling server managing rooms, chat, presence, and WebRTC offer/answer/ICE over a single WebSocket per client; built custom React hooks (useWebSocketClient, useWebRTC); authored an 11-step tutorial selected as top project in SER 421.

Swipe-to-Shop Fashion PWA (Clozyt)

Sep 2025

TypeScript, Next.js, PyTorch, Qdrant

- Launched a swipe-driven shopping PWA with adaptive recommendations under 200 ms p95, embedding products with fashion_clip and querying Qdrant with a dynamic exploit/explore mix; normalized messy CSV catalogs into a clean seven-column schema.

FluxRate — Adaptive Pricing Engine

Aug 2025

Java, Maven, Python, CPS Design

- Built a closed-loop cyber-physical pricing simulator (SRC-based gate controllers, plant model, pricing supervisor) for capacity-constrained resources; validated safety properties (capacity bounds, no double billing) across 24-hour, 50-space scenarios with a Python analysis pipeline.

Market Risk Forecasting Dashboard

Apr 2025

Python, TensorFlow, AWS SageMaker, SQL

- Forecast market volatility with LSTM models, beating GARCH baselines by 16%; built Python + SQL pipelines ingesting 5+ years of OHLCV data (1M+ rows) with engineered features, cutting preprocessing time by 30%.

Safe Python Execution Service

Mar 2025

Python, Flask, nsjail, Docker, GCP Cloud Run

- Designed a Cloud Run microservice executing untrusted Python under nsjail (no network, rlimits, timeouts) with structured JSON results; sub-300 ms p95 latency and scale-to-zero cost profile.

HAAS — Logic-Driven Programming Language

Feb 2025

SWI-Prolog DCGs, Flex/Bison, GCC

- Implemented a full compiler pipeline (tokenizer, parser, AST, runtime evaluator) with 5+ language features and an interactive REPL; tutorials and output redirection cut onboarding time by 30%.

Agile Project Management App

Jan 2025

Java, Maven

- Built an Agile planning app with integrated data-driven Planning Poker estimation, boosting sprint estimation accuracy across 5+ team projects.

Pokemon Game Engine

Dec 2024

C++, OOP, Data Structures

- Developed a modular battle engine validated across 50+ simulated battles, with extensible design for new creatures and abilities.

WhatsApp Group Chat Analysis

Oct 2024

Python, Pandas, NLTK, scikit-learn

- Processed 10,000+ messages to surface sentiment and conversational trends; regex-based preprocessing achieved 98% parsing accuracy.

Senti Chat

Sep 2024

React, Next.js, Express, Pusher, NLTK

- Built a real-time chat app tagging each message with sentiment/emotion via an NLP service (NLTK + scikit-learn); covered with Jest/RTL unit tests.

Crypto Notes DApp

Aug 2024

React, Solidity, Hardhat, IPFS, GraphQL

- Created a token-gated decentralized notes app on Ethereum with IPFS persistence; 90%+ test coverage and ~30% lower query latency via GraphQL.

Image Restoration of Natural Images (MPRNet)

Apr 2021

Python, TensorFlow

- Reduced noise by 45% and improved PSNR by 5.2 dB over baselines using an MPRNet denoising pipeline.

Image Clustering for Script Digitalization

Feb 2020

Python, scikit-learn, KMeans, PCA

- Grouped handwritten scripts with 92% cluster consistency, improving digitalization efficiency by 30%.